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ORIGINAL RESEARCH

Genome-wide identification, characterization and functional prediction of the *SPL* gene family in sesame (*Sesamum indicum* L.)

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Abstract

'SQUAMOSA promoter binding protein-like' (SPL) proteins are the plant-specific transcription factors (TFs) family, comprising a highly conserved SBP domain with two zinc-fingers. It plays a vital role in plant growth and development and stress responses in plants. The SPL TF family is well characterized in several plant species, nevertheless not yet accomplished in sesame (*Sesamum indicum* L.). In this study, we performed in silico identification, molecular characterization, phylogenetic relationship, *cis*-acting regulatory elements, protein-protein interaction, syntenic relationship, duplication events and RNA-seq based expression of sesame *SPL* genes. We identified a total 19 of *SiSPL* genes on 12 different LGs of sesame genome through comparing with the other plant species. The *SiSPL* genes varied in gene structures like 2–11 exons and 1–10 introns. The SiSPL proteins contained a highly conserved 'SQUAMOSA-promoter binding protein' (SBP, pfam03110), only found in plants. Further, SiSPL4 and SiSPL9 contained one extra domain called 'Ank_2' (pfam12796). The *SiSPL* genes were grouped into seven groups via phylogenetic analysis which showed the relationship with other plant species. Total 454 stress responsive '*cis*-regulatory' elements were detected in the promoter region of 19 *SiSPL* genes indicating their possible involvement in stress responses. Furthermore, the available sesame transcriptomic data indicated that *SiSPL* genes varied in transcript accumulation between stressed vs. control samples. In particular, *SiSPL9* (SIN_1016140), *SiSPL12* (SIN_1017992) and *SiSPL3* (SIN_1006254) might be drought responsive. Likewise, *SiSPL8* (SIN_1013914), *SiSPL15* (SIN_1023563) and *SiSPL17* (SIN_1026466) was up-regulated in salinity conditions. Besides, the transcript abundance of *SiSPL1* and *SiSPL9* genes was higher in waterlogging conditions as compared to the control. To best of our knowledge, this is the first report of *SPL* gene family in sesame. The findings of this study will encourage further research into gene expression as well as functional analysis of *SiSPL* genes in sesame and other plant species.

Keywords *SiSPL* gene family · Transcription factor · Characterization · *Sesamum indicum*

Abbreviations

AA	Amino acid
FPKM	Fragments per kilobase of transcript per million mapped reads
GRAVY	Grand average of hydropathicity
h	Hour
Ka	Non-synonymous substitution rate
Ks	Synonymous substitution rate
MW	Molecular weights
LG	Linkage group
pI	Isoelectric points
RPKM	Reads per kilobase of transcript per million mapped reads
SBP	SQUAMOSA-promoter binding protein
SiSPL	Sesamum indicum SQUAMOSA promoter binding protein-like
SSR	Simple sequence repeat
TFs	Transcription factors

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Introduction

The members of *SPL* gene family are the plant specific transcription factors (TFs) consisting of a highly conserved 'SQUAMOSA-promoter binding protein' (SBP) DNA binding domain of 76 aa residues with two 'Zn^{2+} binding' sites ('Cys-Cys-His-Cys' and 'Cys-Cys-Cys-His') (Cao et al. 2019; Yamasaki et al. 2004). The 'Zn^2+'finger sites are found in the 'N-terminal' and 'C-terminal' positions of SBP domain (Birkenbihl et al. 2005; Yamasaki et al. 2004). In addition, the C-terminal region of SBP domain contains a nuclear localization signal (NLS) which overlaps with 'Zn^2+' finger site (Birkenbihl et al. 2005). These genes play a significant role in many plant processes, including plant growth and development, particularly in floral development, transition from vegetative to reproductive phase, leaf initiation, grain size, quality and yield, trichome and fruit development, ripening and signal transduction (Abdullah et al. 2018; Gandikota et al. 2007; Shikata et al. 2009; Singh et al. 2024; Yamasaki et al. 2004; Zhou et al. 2018). Besides, recent studies have also revealed their role in abiotic stress responses, including salinity and drought stresses in several plant species (He et al. 2022; Hou et al. 2018; Mao et al. 2016; Shaheen et al. 2024).

Sesame (*Sesamum indicum* L.) is one of the oldest oil crops and widely cultivated in tropical and subtropical regions of Asia, Africa and South America (Bedigian and Harlan 1986; Sharaby and Butovchenko 2019; Wang et al. 2022). The cultivated sesame is a self-pollinated ($2n = 2x = 26$) species that belongs to the Pedaliaceae family. It is also rich in proteins (about 20%) and various antioxidants (like sesamin, sesamolin, sesamol, episesamin, γ-tocopherol etc.) (Chen et al. 2005; Dalibalta et al. 2020). The seed contains a high oil content (about 55% of the seed weight) compared to oil crops (Wei et al. 2017). Sesame seeds provide edible oil for cooking as well as proteins in the human diet. In addition, its oil is used in the salad and margarine, and pharmaceutical industries (Anilakumar et al. 2010; Dalibalta et al. 2020; Jamieson and Baughman 1924; Salunkhe and Desai 1986). It has potential health benefits to humans as it acts as an antiaging, anticancer, cholesterol lowering, and reduces blood pressure (Anilakumar et al. 2010; Chen et al. 2005; Dalibalta et al. 2020; Wei et al. 2022).

Recently, the sesame genome (approximately 309.35 Mb) has been sequenced (Wang et al. 2022). The availability of sesame genomic information offers a valuable basis for identifying and characterizing the important gene families at the molecular level. Several previous research reports revealed that different members of the *SPL* gene family showed their transcript abundance during abiotic stress conditions. For example, *ZmSPL2* and *ZmSPL20* in maize for cold and drought (Mao et al. 2016), *BpSPL9* in birch for drought and salinity (Ning et al. 2017), *MsSPL8* in alfalfa for drought and salinity (Gou et al. 2018), *CqSPL1* and *CqSPL5* in waterlogging stress (Ren et al. 2022). The potential roles of sesame *SPL* genes in growth, development, and abiotic stress tolerance can be comprehended through identifying and analyzing the expression patterns in different organs and under abiotic stress conditions. Furthermore, the findings this study could shed light on the molecular mechanisms to develop abiotic stress-resilient sesame cultivars with improved agronomic traits. However, to date, a very few gene families are characterized in sesame. This study aimed to conduct in silico identification and characterization of candidate *SPL* genes in sesame. Thus, we analyzed their physicochemical properties, domain and motif composition, syntenic relationships, duplication events, and expression profiles.

Materials and methods

Identification of *SPL* genes in sesame genome

We retrieved the protein sequences of *SPL* gene family from the 'The Arabidopsis Information Resource' (https://www.arabidopsis.org/) and performed the 'BLASTP' search against the sesame genome in 'Ensembl Plants' (https://plants.ensembl.org/index.html) database. Then again, we retrieved the SPL protein sequences of other previously reported plant species, including *Triticum aestivum* (Li et al. 2022), *Capsicum annum* (Zhang et al. 2016), *Glycine max* (Tripathi et al. 2017), *Helianthus annuus* (Jadhao et al. 2023) and *Brassica napus* (Cheng et al. 2016), and performed 'BLASTP' search to identify candidate *SPL* genes in sesame genome. Moreover, the putative *SPL* genes in sesame were re-checked that may contain 'SQUAMOSA-promoter binding protein' (SBP, 'pfam03110') domain by 'HMMER' web server that uses the profile 'hidden Markov Models (HMM)' (Potter et al. 2018) and 'Pfam' (http://pfam.xfam.org) database.

Phylogenetic analysis and protein sequence alignment of *SiSPL* genes

A phylogenetic analysis was carried out with the protein sequences *SPL* genes of *S. indicum*, *(A) thaliana*, and *(B) napus* by using MEGA11 (Tamura et al. 2021) to comprehend the evolutionary relationship of *S. indicum SPL* genes with other plant species. The protein sequences of *SPL* genes of these species were aligned with 'ClustalW' and a circular phylogenetic tree was made following the 'Neighbor-Joining (NJ)' method with 1000 bootstrap using the default parameters. After that, the tree was visualized with an online visualization tool 'iTOL', freely available at 'https://itol.embl.de/'.

Molecular characterization of *SiSPL* genes

The nucleotide sequence length (bp), amino acid sequence length (aa) as well as the chromosomal position of the *SiSPL* genes were anticipated from 'Ensembl Plants'. The 'molecular weight' (MW), 'isoelectric point' (pI) and 'grand average of hydropathicity' (GRAVY) of SiSPL proteins were analyzed with the 'ProtParam tool' (https://web.expasy.org/protparam/). The 'Plant-mPLoc' (an online tool available at 'http://www.csbio.sjtu.edu.cn/bioinf/plant-multi/') was used to identify the subcellular localization of *SiSPL* genes. The gene structure (exon-intron) of *SiSPL* genes was determined by the 'Gene Structure Display Server (GSDS2.0)'(Hu et al. 2015). Likewise, the 'conserved domains' and 'motifs' of SiSPL proteins were predicted with the 'CD-search' tool of 'Conserved Domain Database (CDD)' available at 'https://www.ncbi.nlm.nih.gov/Structure/cdd/wrpsb.cgi' and 'MEME' available at 'https://meme-suite.org/tools/meme', respectively. The genomic location of *SiSPL* genes was performed with 'MapChart' (Voorrips 2002). Moreover, the '3D structure' of SiSPL proteins was predicted with the 'SWISS-MODEL' available at 'https://swissmodel.expasy.org/interactive#sequence'. The '*cis*-regulatory elements' of *SiSPL* genes were predicted using 'PlantCARE' available at 'https://bioinformatics.psb.ugent.be/webtools/plantcare/html/' within the 3 Kb upstream region of the transcription start site. The protein-protein interaction network was determined by 'STRING', available at 'https://string-db.org/'. The synteny relationship of *SiSPL* genes was determined by using the 'Circoletto' tool (Darzentas 2010). For gene duplication analysis, the Ka and Ks values, and Ka/Ks ratio for *SiSPL* pairs were calculated with 'TBtools' package (Yang and Nielsen 2000). The gene ontology (GO) enrichment analysis of *SiSPL* genes was performed with the 'g: Profiler' (Kolberg et al. 2023).

Genetic variation, transcriptome data and expression pattern of *SiSPL* genes

The genetic variation like simple sequence repeat (SSR) present in *SiSPL* genes was identified by the 'Web SSR Finder' tool of 'Plant Simple Sequence Repeat Database (PSSRD)' available at 'http://www.pssrd.info/' (Song et al. 2021). The RNA-sequencing data (FPKM expression) was retrieved from the 'Gene Expression Omnibus (GEO)' database of NCBI under the accession numbers 'GSE164511' (young leaf), and 'GSE148340' (control vs PEG-treated root) of sesame. Likewise, the RPKM expression data of sesame shoot under salinity conditions (tolerant vs sensitive) was used from published articles (Zhang et al. 2020). Besides, the FPKM value of water logging and control root (GSE133186) of sesame at flowering stage was also used from previous research report (Dossa et al. 2019). Finally, the heatmap was generated using the 'TBtools' package (Yang and Nielsen 2000).

Results

Identification of putative *SPL* genes in sesame genome

The members of the sesame (*S. indicum) SPL* (*SiSPL*) gene family were identified by the 'BLASTP' search with the protein sequences of *AtSPL* genes against the sesame genome in 'Ensembl Plants' database (https://plants.ensembl.org/index.html). Then again, protein sequences were also 'BLASTP' searched and cross-checked. Finally, a total of 19 unique *SiSPL* genes were identified in sesame genome (Tables 1 and Supplementary file 1). Furthermore, the protein sequences were further confirmed with 'HMMER' and 'Pfam'.

Phylogenetic analysis of *SiSPL* genes

A circular phylogenetic tree was constructed with SPL proteins of *S. indicum*, *(A) thaliana*, *T. aestivum*, *C. annum*, *G. max*, *H. annuus* and *(B) napus* using MEGA11 to determine the evolutionary relationships (Fig. 1 and Supplementary file 1). The result revealed that 19 *SiSPL* genes were grouped into seven groups (Fig. 1). Group IV had a maximum five *SiSPL* genes like *SiSPL8* (SIN_1013914), *SiSPL12* (SIN_1017992), *SiSPL16* (SIN_1024610), *SiSPL18* (SIN_1026481) and *SiSPL19* (SIN_1026483) while group I and II contained only two *SiSPL* genes *SiSPL7* (SIN_1013913) and *SiSPL9* (SIN_1016140), respectively. Group III had three *SiSPL* genes, including *SiSPL2* (SIN_1005293), *SiSPL11* (SIN_1016944) and *SiSPL17* (SIN_1026466), and group V contained *SiSPL3* (SIN_1006254) and *SiSPL15* (SIN_1023563). Group VI exhibited four *SiSPL* genes, such as *SiSPL1* (SIN_1004240), *SiSPL4* (SIN_1007413), *SiSPL5* (SIN_1009415) and *SiSPL13* (SIN_1018251) and group VII contained three *SiSPL* genes like *SiSPL6* (SIN_1010138), *SiSPL10* (SIN_1016272) and *SiSPL14* (SIN_1022592).

Physicochemical properties, structure, conserved domain and motif analysis of *SiSPL* genes

The various physicochemical characteristics of 19 *SiSPL* genes, including gene sequence length (bp), protein size (aa), 'theoretical isoelectric point' (pI), 'molecular weight'

Table 1 Characteristics of members of the *SPL* gene family in *S. indicum*

Sl. No.	Gene name	Gene ID	Location	Gene length (bp)	Protein length (aa)	Sub-cellular localization	Molecular weight (kDa)	Isoelectric point (pI)	Gravy	No. of exons
1	*SiSPL1*	SIN_1004240	LG7:988781.994993	6213	1075	Cytoplasm/nucleus	118.89	6.63	-0.417	10
2	*SiSPL2*	SIN_1005293	LG12:4356458.4357693	1236	131	Nucleus	15.01	8.57	-1.250	2
3	*SiSPL3*	SIN_1006254	LG4:568554.572311	3758	369	Nucleus	39.22	9.05	-0.664	3
4	*SiSPL4*	SIN_1007413	LG2:2786671.2793507	6837	1080	Cytoplasm/nucleus	119.98	7.19	-0.434	10
5	*SiSPL5*	SIN_1009415	LG1:10710598.10712836	2239	463	Nucleus	51.07	7.24	-0.587	3
6	*SiSPL6*	SIN_1010138	LG10:1067472.1073524	6053	775	Cytoplasm/nucleus	86.43	7.13	-0.358	10
7	*SiSPL7*	SIN_1013913	LG1:16663084.16664646	1563	326	Cytoplasm/nucleus	36.14	8.66	-0.781	3
8	*SiSPL8*	SIN_1013914	LG1:16667457.16669355	1899	389	Nucleus	43.04	8.68	-0.672	3
9	*SiSPL9*	SIN_1016140	LG3:116742.122104	5363	935	Nucleus	103.56	6.40	-0.391	11
10	*SiSPL10*	SIN_1016272	LG3:1096712.1104136	7425	786	Cytoplasm/nucleus	87.80	6.61	-0.299	10
11	*SiSPL11*	SIN_1016944	LG16:1934422.1936323	1902	168	Nucleus	17.96	9.46	-0.987	2
12	*SiSPL12*	SIN_1017992	LG2:18354696.18357863	3168	470	Nucleus	51.29	8.66	-0.630	4
13	*SiSPL13*	SIN_1018251	LG2:16834490.16836656	2167	462	Nucleus	50.87	7.22	-0.470	3
14	*SiSPL14*	SIN_1022592	LG6:21625264.21626899	1636	327	Nucleus	37.09	8.47	-0.911	4
15	*SiSPL15*	SIN_1023563	LG5:12812691.12815124	2434	356	Nucleus	38.01	9.11	-0.639	3
16	*SiSPL16*	SIN_1024610	LG11:9841161.9842349	1189	327	Nucleus	35.97	8.25	-0.557	3
17	*SiSPL17*	SIN_1026466	LG8:12894257.12896422	2166	305	Cytoplasm/nucleus	33.48	7.60	-0.583	2
18	*SiSPL18*	SIN_1026481	LG8:12753506.12755968	2463	469	Cytoplasm/nucleus	51.16	7.25	-0.562	4
19	*SiSPL19*	SIN_1026483	LG8:12735022.12737258	2237	470	Cytoplasm/nucleus	51.20	8.93	-0.622	4

(MW), 'grand average of hydropathicity' (GRAVY) and 'subcellular localization' are shown in Table 1. The length of *SiSPL* genes varied from 1236 to 6837 bp while protein length ranged from 131 to 1080 aa. The *SiSPL2* (SIN_1005293) and *SiSPL4* (SIN_1007413) genes showed the maximum (131 aa) and minimum (1080 aa) protein lengths, respectively. The *SiSPL9* (SIN_1016140) and *SiSPL11* (SIN_1016944) showed maximum (6.40) and minimum (9.46) 'isoelectric point' (pI). The molecular weight of SiSPL proteins ranged from 15.01 kDa (SiSPL2) to 119.98 kDa (SiSPL4). Similarly, the SiSPL proteins showed negative 'GRAVY' values which were ranging from − 1.25 (SiSPL2) to -0.299 (SiSPL10) demonstrating the hydrophilic activities of SiSPL proteins. The subcellular localization analysis revealed that 11 SiSPL proteins were located in the nucleus, including SiSPL2 (SIN_1005293), SiSPL3 (SIN_1006254), SiSPL5 (SIN_1009415), SiSPL8 (SIN_1013914), SiSPL9 (SIN_1016140), SiSPL11 (SIN_1016944), SiSPL12 (SIN_1017992), SiSPL13 (SIN_1018251), SiSPL14 (SIN_1022592), SiSPL15 (SIN_1023563) and SiSPL16 (SIN_1024610) whereas other 8 SiSPL proteins like SiSPL1 (SIN_1004240), SiSPL4 (SIN_1007413), SiSPL6 (SIN_1010138), SiSPL7 (SIN_1013913), SiSPL10 (SIN_1016272), SiSPL17 (SIN_1026466), SiSPL18 (SIN_1026481) and SiSPL19 (SIN_1026483) were located in both nucleus and chloroplast. The structure (exon–intron) of *SiSPL* genes was predicted using 'GSDS2.0' program and the result showed that their exon number varied from 2 to 11 (Fig. 2; Table 1). The *SiSPL9* (SIN_1016140) had the highest (11) while and *SiSPL2* (SIN_1005293), *SiSPL11* (SIN_1016944) and *SiSPL17* (SIN_1026466) exhibited the lowest (2) number of exons. Likewise, the intron number also differed from 1 to 10 in *SiSPL* genes (Fig. 2). Furthermore, the presence of 'conserved domains' and 'motifs' in SiSPL proteins was analyzed using 'CD-search' tool of 'NCBI' and 'MEME' suite (Figs. 3, 4 and 5; Tables 2 and 3). The result indicated that all the SiSPL proteins had a highly conserved 'SQUAMOSA-promoter binding protein' (SBP, pfam03110) found in plants. Besides, SiSPL4 and SiSPL9 contained one additional domain called 'Ank_2' (pfam12796) with three copies of Ankyrin repeats. Likewise, 15 conserved 'motifs' with over 25 aa residues in SiSPL proteins (Fig. 4 and Table 3). Of these motifs, motif 1 and 2 were predominantly conserved to 'SBP' domains for all SiSPL proteins (Fig. 5). Additionally, we aligned the protein sequences and found that the conserved amino acid residues in the 'SBP' domains and motif 1 for all 19 *SiSPL* genes (Fig. 6).

Structure of SiSPL proteins and protein-protein interactions

For each of the 19 SiSPL proteins, a three-dimensional structure was predicted (Fig. 7) using an online tool "SWISS-MODEL". The findings demonstrated that each SiSPL protein has a disorganized structure. Upon examining the structures, it was observed that the SiSPL1 (SIN_1004240) and SiSPL4 (SIN_1007413), proteins

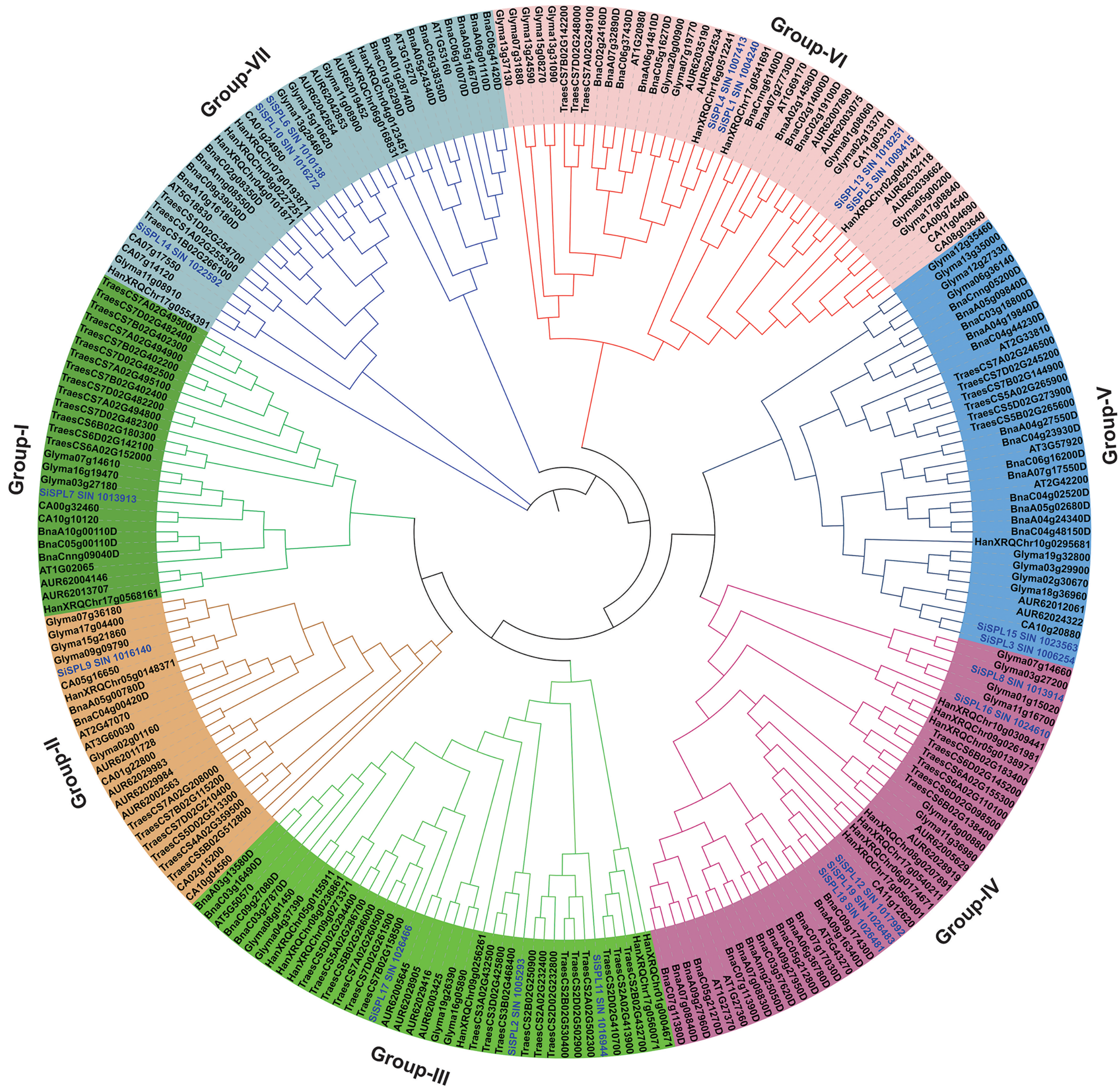

Fig. 1 The circular representation of phylogenetic tree SPL proteins in *S. indicum*. The phylogenetic analysis was carried out the SPL protein sequences of different plant species, including *S. indicum*, *Arabidopsis thaliana*, *Triticum aestivum*, *Capsicum annum*, *Glycin max*, *Helianthus annuus* and *Brassica napus*. The tree was built using MEGA11 considering 'Neighbor-Joining (NJ)' method

shared a nearly identical structure with a profusion of α and β-sheets. However, in the case of SiSPL2 (SIN_1005293) and SiSPL11 (SIN_1016944), both α-helix and β-sheets were vaguely visible. Moreover, the proportion of β-sheet in SiSPL3 (SIN_1006254), SiSPL5 (SIN_1009415), SiSPL7 (SIN_1013913), and SiSPL8 (SIN_1013914) is significantly higher than that of α-helix. Conversely, SiSPL6 (SIN_1010138), SiSPL9 (SIN_1016140), and SiSPL10 (SIN_1016272) have a cluster of α-helices and very few β-sheets. On the other hand, in SiSPL15 (SIN_1023563), the α-helix was completely absent. The remaining structures had noticeable β-sheets, and not so precise α-helices. Moreover, the protein-protein interaction network for SiSPL proteins was constructed (Fig. 8) using the *A. thaliana* protein information (Supplementary file 2) using an online tool 'STRING' (https://string-db.org/). According to the analysis, SPL7 directly interacts with COPT1, COPT2 and COPT6. Six different SPL proteins, including SPL4, SPL6, SPL9, SPL10, SPL11, and SPL13B have a robust interaction with TOE2.

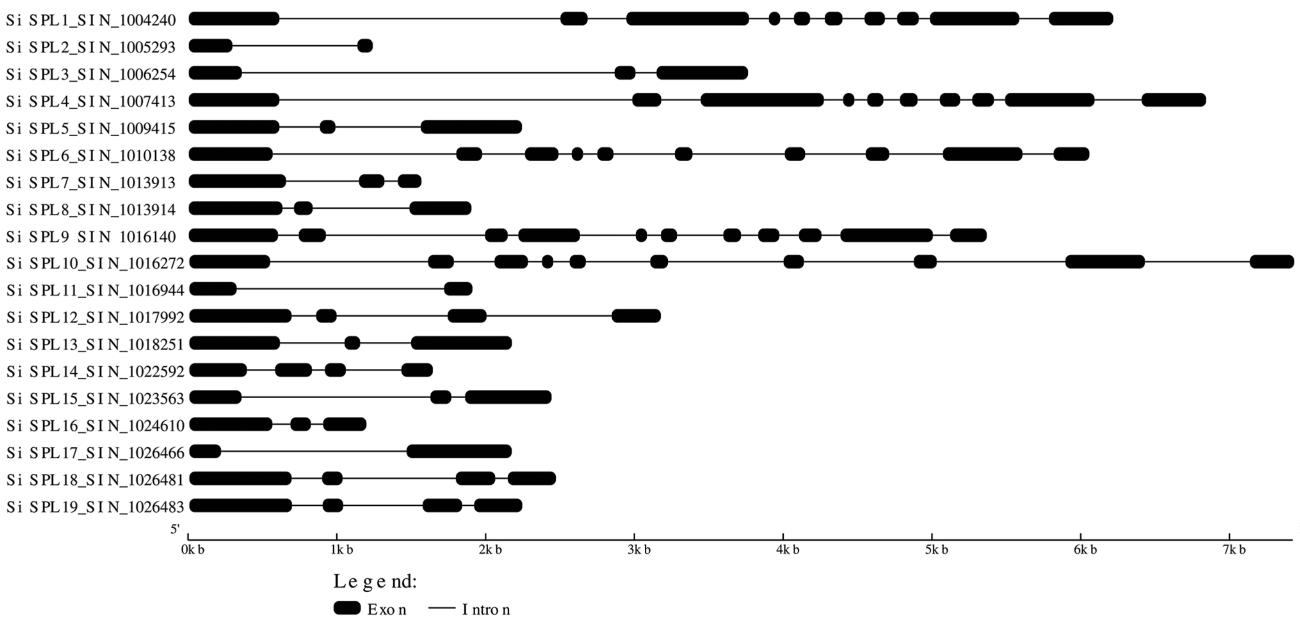

Fig. 2 The gene (exon-intron) structure of *SiSPL* genes

Position of *SiSPL* genes in the sesame genome

The genomic position of 19 *SiSPL* genes was examined (Table 1) and visualized with the 'MapChart' program. The result indicated that 19 *SiSPL* genes were located in 12 different 'linkage groups' (LGs). Of which, *SiSPL5*, *SiSPL7* and *SiSPL8* were located in LG1, *SiSPL4*, *SiSPL12* and *SiSPL13* in LG2, *SiSPL9* and *SiSPL10* in LG3, *SiSPL3* in LG4, *SiSPL15* in LG5, *SiSPL14* in LG6, *SiSPL1* in LG7, *SiSPL17*, *SiSPL18* and *SiSPL19* in LG8, *SiSPL6* in LG10, *SiSPL16* in LG11, *SiSPL2* in LG12, and *SiSPL11* in LG16 (Fig. 9).

Syntenic relationship and gene duplication analysis of *SiSPL* genes

The orthologous relationship and evolutionary conservation of *SiSPL* genes with *(A) thaliana*, *T. aestivum*, *C. annum*, *G. max*, *H. annuus* and *(B) napus*. *G. max*, and *O. sativa* were determined by a web-based tool 'Circoletto' (Fig. 10 and Supplementary file 3). The outcome of the analysis indicated that 19 *SiSPL* genes had syntenic relationship with 10 genes in *(A) thaliana*, 10 genes in *C. annum*, 13 genes in *T. aestivum*, 13 genes in *G. max*, 12 genes in *H. annuus* and 13 genes in *(B) napus*. Moreover, the gene duplication events for *SiSPL* genes were explored. We identified five pairs of genes that exhibited high similarity and fall within the same clade in the phylogenetic tree (Fig. 1) suggesting that there is a probability of gene duplication events for these five pairs of *SiSPL* genes. Consequently, the 'BLASTP' as 'all-vs-all' (*SiSPL1*, *SiSPL3*, *SiSPL4*, *SiSPL6*, *SiSPL10*, *SiSPL13*, *SiSPL15*, *SiSPL18 and SiSPL19*) was performed using 'TBtools' and considered the threshold e-value less than $1 \times e^{-152}$ (Supplementary file 4). The result showed that *SiSPL1* and *SiSPL4*, *SiSPL3* and *SiSPL15*, *SiSPL5* and *SiSPL13*, *SiSPL6* and *SiSPL10*, and *SiSPL18* and *SiSPL19* could be duplicated genes. Further, the 'non-synonymous divergence' (Ka) and 'synonymous divergence' (Ks) and Ka/Ks ratio of gene pairs were determined using the 'TBtools' program (Table 4). Generally, the ratio of Ka/Ks < 1, Ka/Ks = 1 and Ka/Ks > 1 represents purifying, neutral and positive selections, respectively (Nekrutenko et al. 2002). The Ks/Ks ratio of *SiSPL* gene pairs varied from 0.2374 to 0.3324 indicating that a robust purifying selection force in the evolution of *SiSPL* proteins.

Analysis of cis-acting elements in the promoter region of *SiSPL* genes

The promoter sequences (3 Kb upstream) of 19 *SiSPL* genes were retrieved from the 'Ensembl Plants' database (Supplementary file 5). We identified a total of 918 '*cis*-regulatory' elements in the promoter region of *SiSPL* genes (Table 5) with 'PlantCARE'. Three categories have been established for the entire promoter region: light response, phytohormone response, and stress response (Zhou et al. 2023). Among these, 183 are associated with phytohormone response (ABRE, P-box, TATC-box, ERE, and TGACG-motif), 454 are related to stress response (DRE CORE, WRE3, WUN-motif, ARE and box S), and 281 are related to light response (AE-box, Box-4, G-Box, and GATA motif). Of which, all *SiSPL* genes included Box-4 (which is involved in light

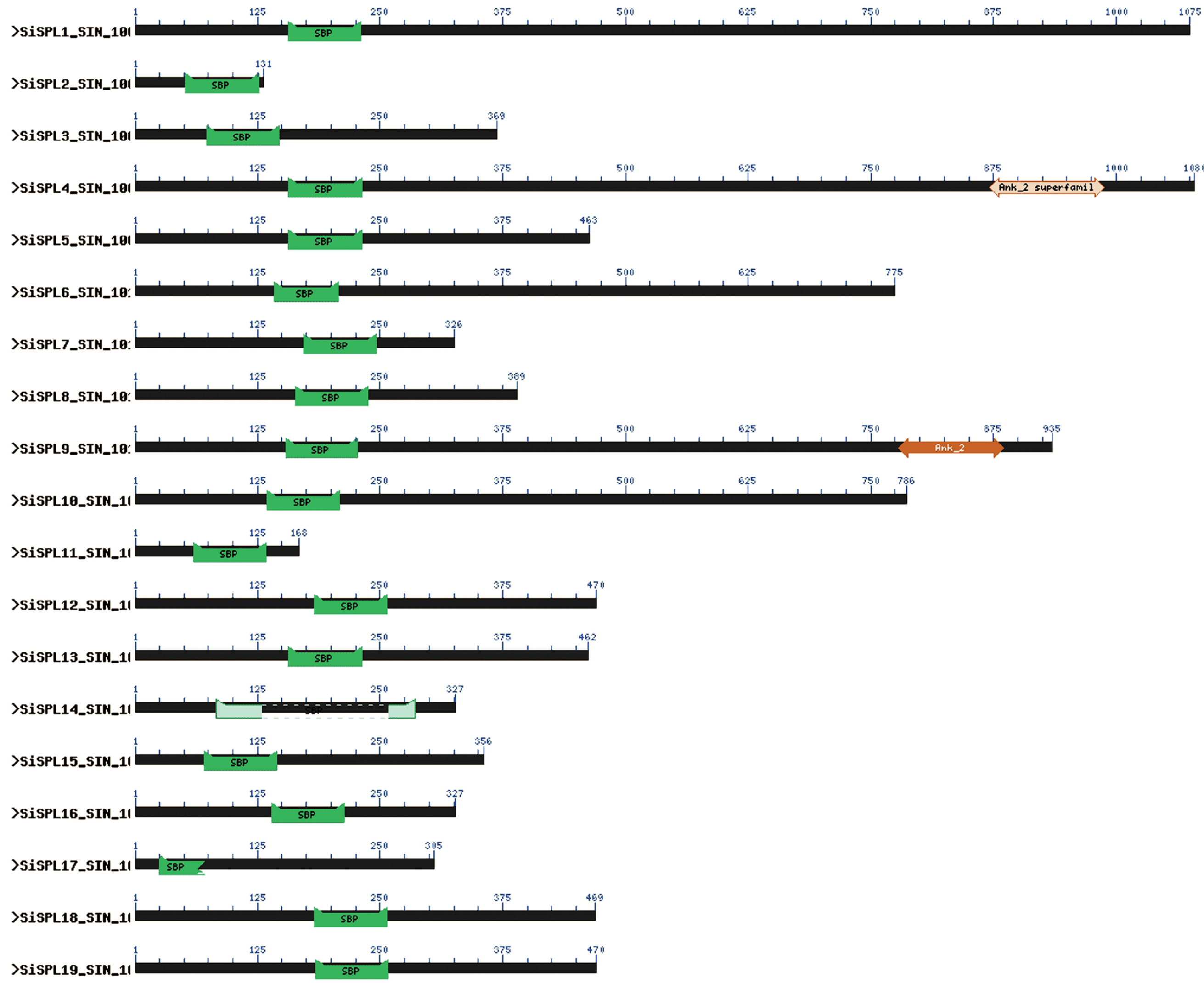

Fig. 3 The domain structure of SPL proteins of *S. indicum*. The domains were anticipated by CD-search of NCBI (https://www.ncbi.nlm.nih.gov/Structure/bwrpsb/bwrpsb.cgi). The domain information are shown in Table 2. Domain names are labeled in the diagram of each protein

response), STRE, and MYB (which are both involved in stress response). However, the promoter region of *SiSPL10* (SIN_1016272) had the greatest number of motifs (75) and the highest number of BOX-4 motifs (13) among 19 *SiSPL* genes. The *SiSPL16* (SIN_1024610), in particular, had the greatest number of LTRs (4), indicating a low sensitivity to temperature.

GO annotation of SiSPL proteins

The functions of 19 SiSPL proteins were categorized into biological processes, molecular functions, and cellular components using ‘g: Profiler’ tool (Fig. 11). Among these, the molecular functions (MF) category, the significantly enriched proteins were DNA binding (GO:0003677) and metal ion binding (GO:0046872) while regulation of vegetative phase change (GO:0010321) and inflorescence development (GO:0010229) were enriched term for biological process (BP) category. As regards the cellular components (CC) category, the nucleus (GO:0005634) was mostly enriched term.

Genetic variation and expression pattern of *SiSPL* genes with transcriptomic data

We analyzed the simple sequence repeat (SSR) motifs in the genomic sequence of 19 *SiSPL* genes (Table 6). The results showed that only seven *SiSPL* genes, including *SiSPL1* (SIN_1004240), *SiSPL4* (SIN_1007413), *SiSPL6* (SIN_1010138), *SiSPL9* (SIN_1016140), *SiSPL14* (SIN_1022592), SiSPL15 (SIN_1023563) and *SiSPL17* (SIN_1026466) exhibited SSR motifs. A/T, TA/TA, TC/GA and AGG/CCT were the most frequent SSR motifs ranging from mono- to tri-nucleotides in the *SiSPL* genes. However,

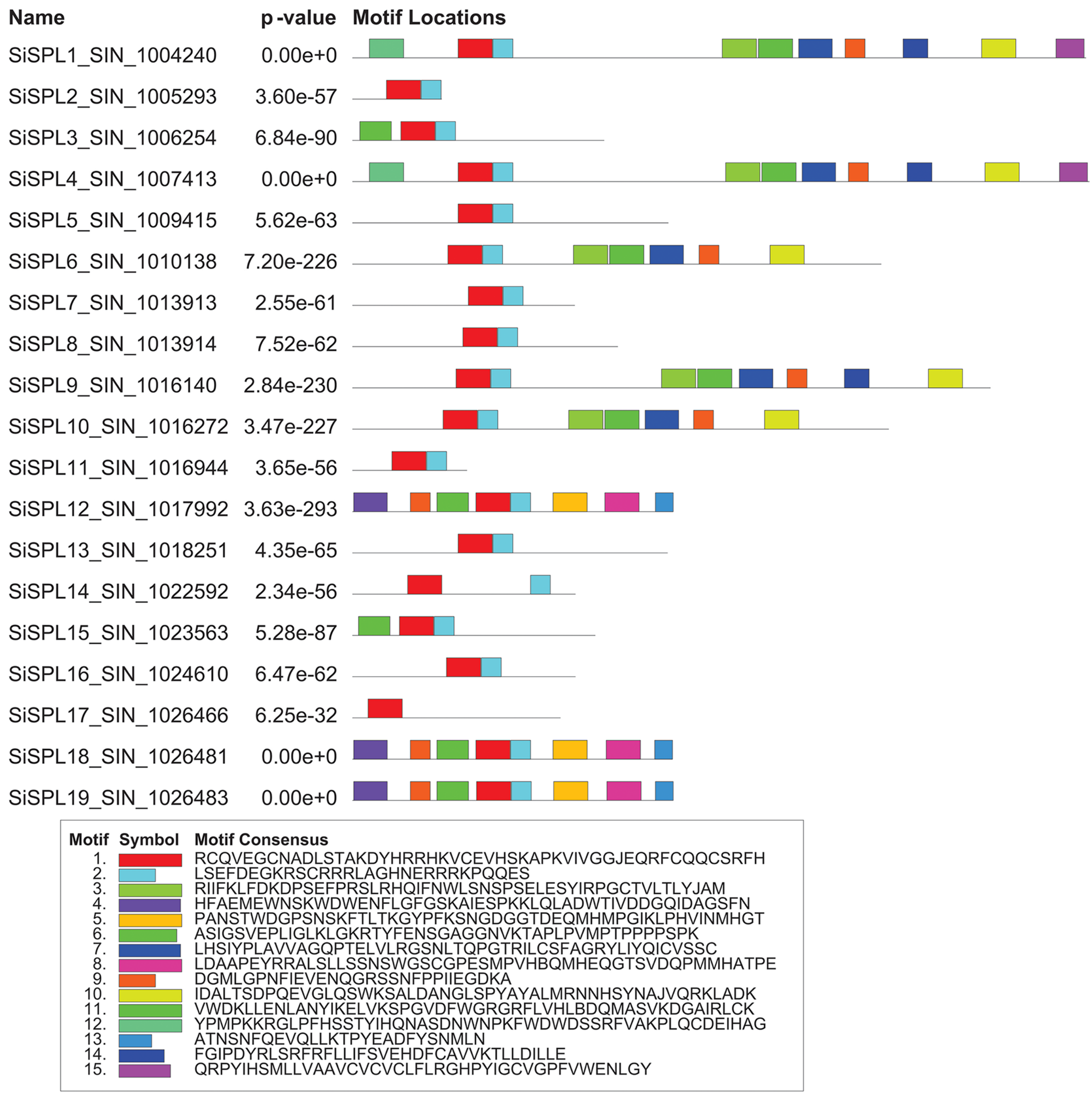

Fig. 4 The conserved motifs of present in 19 SPL proteins of *S. indicum*. Motifs are labelled with different colored rectangles from 1 to 15. Additional information is given in Table 3

SiSPL4 (SIN_1007413), *SiSPL6* (SIN_1010138), *SiSPL14* (SIN_1022592) and *SiSPL17* (SIN_1026466) genes had tri-nucleotide SSR motifs with repeat size 15, 21, 21 and 15 bp, respectively.

We checked the in silico expression pattern of *SiSPL* genes in various tissues of sesame, including leaf, root (under drought), shoot (under salinity) with the transcriptome based 'FPKM'/'RPKM' datasets obtained from 'gene expression omnibus' (GEO) datasets of the NCBI and previously reported articles (Fig. 12). The results revealed that all 19 *SiSPL* genes were expressed in young leaves of sesame, nonetheless, the expression of *SiSPL1* (SIN_1004240) was followed by *SiSPL9* (SIN_1016140), *SiSPL4* (SIN_1007413), *SiSPL14* (SIN_1022592), *SiSPL6* (SIN_1010138) and *SiSPL18* (SIN_1026481) genes was higher than the other genes (Fig. 12a). Likewise, these *SiSPL* genes were filtered with the FPKM expression values of root transcriptome between control and drought (PEG-treated)

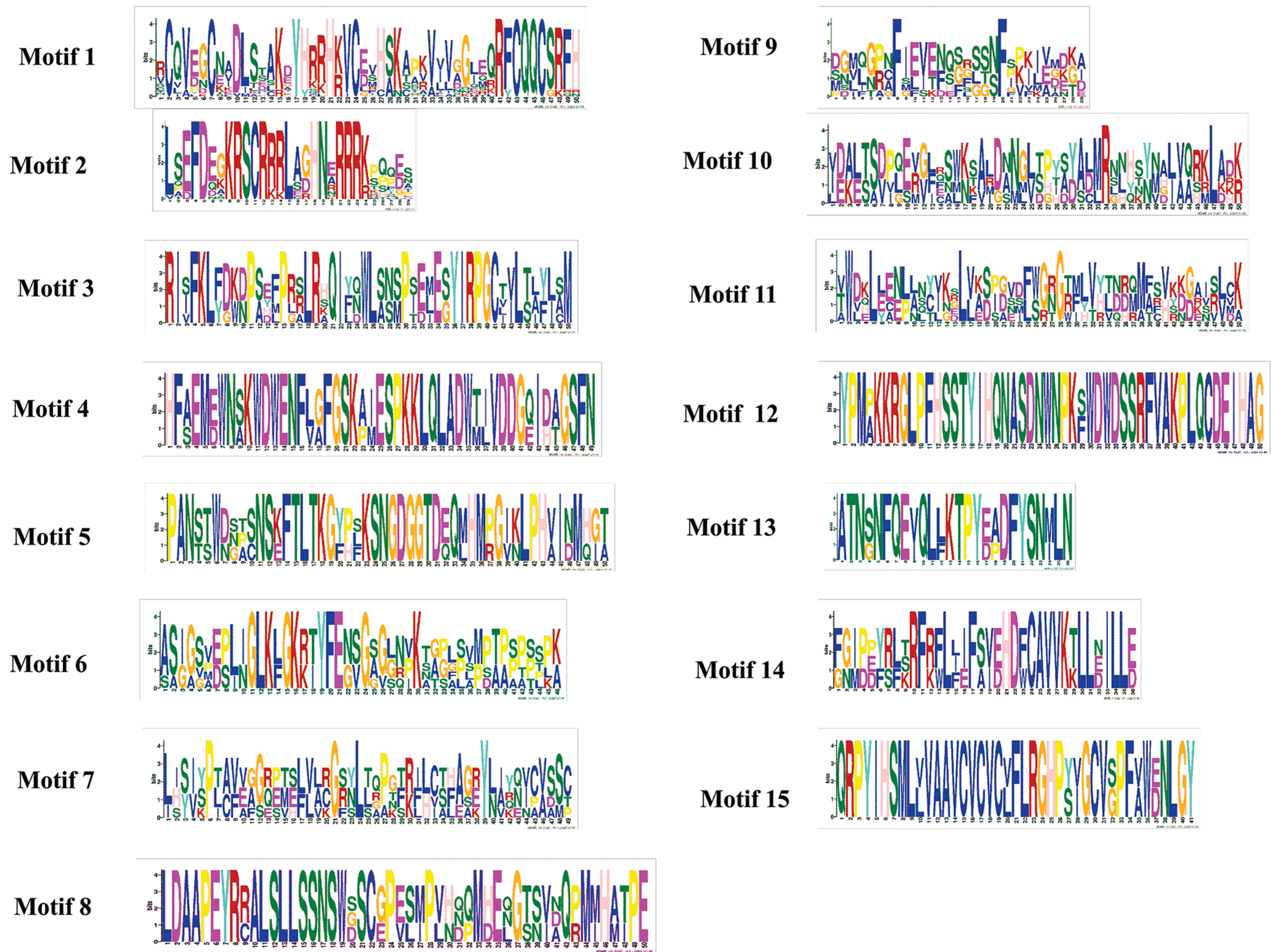

Fig. 5 Logos of the motif 1 and motif 15 present in the 'SBP' and 'Ank_2' domain of SPL proteins in *S. indicum*

Table 2 Major domains present in the 19 *SPL* proteins of *S. indicum*

Sl. No.	Domain name	Description	Functions	Reference
1	SBP	SQUAMOSA-promoter binding protein (SBP), a sequence specific DNA-binding domain which contained 10 conserved cysteine and histidine residues, and found in plants	It involves in the regulation of early flower development	(Klein et al. 1996)
2	AnK_2	Three copies of Ankyrin repeats	Important for cell growth, development, and hormones and stresses responses	(J.-Y. Zhao et al. 2020)

condition (Fig. 12b). The results showed that the majority of *SiSPL* genes were expressed in both control and PEG-treated roots of 'VEN-1' and 'TEX-1' genotypes of sesame except *SiSPL8* (SIN_1013914), *SiSPL11* (SIN_1016944), *SiSPL15* (SIN_1023563) and *SiSPL16* (SIN_1024610) that were hardly detected in few replicates. Nonetheless, the expression of *SiSPL9* (SIN_1016140) followed by *SiSPL12* (SIN_1017992) and *SiSPL3* (SIN_1006254) genes was higher in drought conditions than the control root of both sesame genotypes. Conversely, the expression of *SiSPL4* (SIN_1007413) and *SiSPL17* (SIN_1026466) genes were decreased in drought conditions than the control root of both sesame genotypes. The *SiSPL* genes were analyzed using the transcriptomic data as previously described by (Zhang et al. 2020) under salinity stress in two sesame genotypes. Among 19 *SiSPL* genes, 12 genes were differentially expressed between control and salinity stress in shoot in both salinity tolerant (ST) and salinity sensitive (SS) sesame genotypes (Fig. 12c). Of which, the expression of *SiSPL2* (SIN_1005293) and *SiSPL16* (SIN_1024610) was up-regulated in ST (24 h/0 h) and down-regulated in SS (12 h/0 h). Likewise, the expression of *SiSPL3* (SIN_1006254) and *SiSPL14* (SIN_1022592) was up-regulated in ST (12 h/0 h) but down-regulated in SS (24 h/0 h) and SS (12 h/0 h),

Table 3 The description of the putative conserved motifs found in 19 SPL proteins of *S. indicum*

Motif name	E-value	Sites	Width	Motif sequence
Motif 1	1.8E-581	19	50	RCQVEGCNADLSTAK-DYHRRHKVCEVHSKAP-KVIVGGJEQRFCQQCSRFH
Motif 2	3.20E-294	18	29	LSEFDEGKRSCRRRLAGH-NERRRKPQQES
Motif 3	4.00E-89	5	50	RIIFKLFDKDPSEFPRSL-RHQIFNWLSNSPSELESY-IRPGCTVLTLYJAM
Motif 4	1.40E-52	3	49	HFAEMEWNSKWDWEN-FLGFGSKAIESPKKLQLAD-WTIVDDGQIDAGSFN
Motif 5	1.10E-37	3	50	PANSTWDGPSNSKFTLTK-GYPFKSNGDGGTDEQM-HMPGIKLPHVINMHGT
Motif 6	8.30E-34	5	46	ASIGSVEPLIGLKLG-KRTYFENSGAGGNVKTA-PLPVMPTPPPPSPK
Motif 7	2.50E-26	5	49	LHSIYPLAVVAGQPTELV-LRGSNLTQPGTRILCSFA-GRYLIYQICVSSC
Motif 8	6.40E-31	3	50	LDAAPEYRRALSLLSSN-SWGSCGPESMPVHBQM-HEQGTSVDQPMMHATPE
Motif 9	7.30E-25	8	29	DGMLGPNFIEVENQGRSS-NFPPIIEGDKA
Motif 10	5.30E-27	5	50	IDALTSDPQEVGLQSWK-SALDANGLSPYAYALM-RNNHSYNAJVQRKLADK
Motif 11	9.30E-28	5	50	VWDKLLENLANYIKELVK-SPGVDFWGRGRFLVHLB-DQMASVKDGAIRLCK
Motif 12	6.40E-24	2	50	YPMPKKRGLPFHSS-TYIHQNASDNWNPKFWD-WDSSRFVAKPLQCDEIHAG
Motif 13	4.40E-22	3	26	ATNSNFQEVQLLKTPY-EADFYSNMLN
Motif 14	2.20E-16	3	36	FGIPDYRLSRFRFLLIFSVE-HDFCAVVKTLLDILLE
Motif 15	7.70E-15	2	41	QRPYIHSMLLVAAVCVCV-CLFLRGHPYIGCVGP-FVWENLGY

respectively. Nonetheless, the expression of *SiSPL8* (SIN_1013914), *SiSPL15* (SIN_1023563) and *SiSPL17* (SIN_1026466) was up-regulated in both genotypes at both time points. We further checked the expression of *SiSPL* genes in the root of waterlogging over time and control from two genotypes (R2G and EG) at the flowering stage in sesame using the transcriptome dataset (FPKM value) as previously reported by (Dossa et al. 2019). The expression of *SiSPL1* (SIN_1004240) and *SiSPL9* (SIN_1016140) was higher among *SiSPL* genes, while others in sesame roots under waterlogging conditions (Fig. 12d). Nonetheless, the expression of these two genes was initially lower (0 h) and decreased at 12, 24 and 36 h after waterlogging condition, then again decreased at drainage condition (38 and 48 h) in the roots of both sesame genotypes. The expression of *SiSPL1* and *SiSPL9* genes was lower in control root as compared to the root with waterlogging conditions. Thus suggesting that these genes might be waterlogging stress-responsive.

Discussion

The 'SQUAMOSA promoter binding protein-like' (SPL) proteins are one of the key transcription factors (TFs) in plants (Yamasaki et al. 2004). They are playing an important role in many plant processes, particularly in floral development, transition from the vegetative to reproductive phase, leaf initiation, grain size, quality and yield, trichome development, and fruit development, and ripening, (Abdullah et al. 2018; Gandikota et al. 2007; Shikata et al. 2009; Yamasaki et al. 2004; Zhou et al. 2018). Besides, recent studies have also revealed their role in abiotic stress responses, including salinity and drought stresses in various plants (He et al. 2022; Hou et al. 2018; Mao et al. 2016; Shaheen et al. 2024). The *SPL* gene family is well characterized in *Arabidopsis* and many other agricultural crops. However, *SPL* gene family has never been reported in sesame. Therefore, in this study, we have focused on in silico identification and characterization of candidate *SPL* genes in sesame.

In the past, many research groups reported different numbers of *SPL* genes in several species, like 16 genes in *Arabidopsis* (Cardon et al. 1999), 15 genes in capsicum (Zhang et al. 2016), 43 genes in soybean (Tripathi et al. 2017), 19 gene in rice (Li et al. 2022), 58 gene in wheat (Li et al. 2022), 25 genes in sunflower (Jadhao et al. 2023), 15 genes in tomato (Salinas et al. 2012), 58 in *B. napus* (Cheng et al. 2016) and 23 genes in quinoa (Zhao et al. 2022). In this study, a total of 19 *SiSPL* genes were identified that were located on 12 different linkage groups (LGs) in the sesame genome, including three genes in LG1, LG2 and LG8, two genes in LG3 and only one gene in LG4, LG5, LG6, LG7, LG10, LG11, LG12 and LG16 (Table 1; Fig. 9). The number of *SiSPL* genes (19) identified in this study is consistent with previous reports. The phylogenetic analysis revealed that 19 *SiSPL* genes were categorized into seven groups (Fig. 1) and showed relationships with the *SPL* genes of *(A) thaliana*, *T. aestivum*, *C. annum*, *G. max*, *H. annuus* and *(B) napus* indicating their conservation throughout evolution. Group IV had a maximum of five *SiSPL* genes whereas groups I and II contained only one *SiSPL* gene in each. A similar grouping for *SPL* genes has been shown in sunflower where a total of 59 *SPL* genes grouped into seven groups (Jadhao et al. 2023). However, eight groups of *SPL* genes were also reported in different plant species, including 16 genes in 8

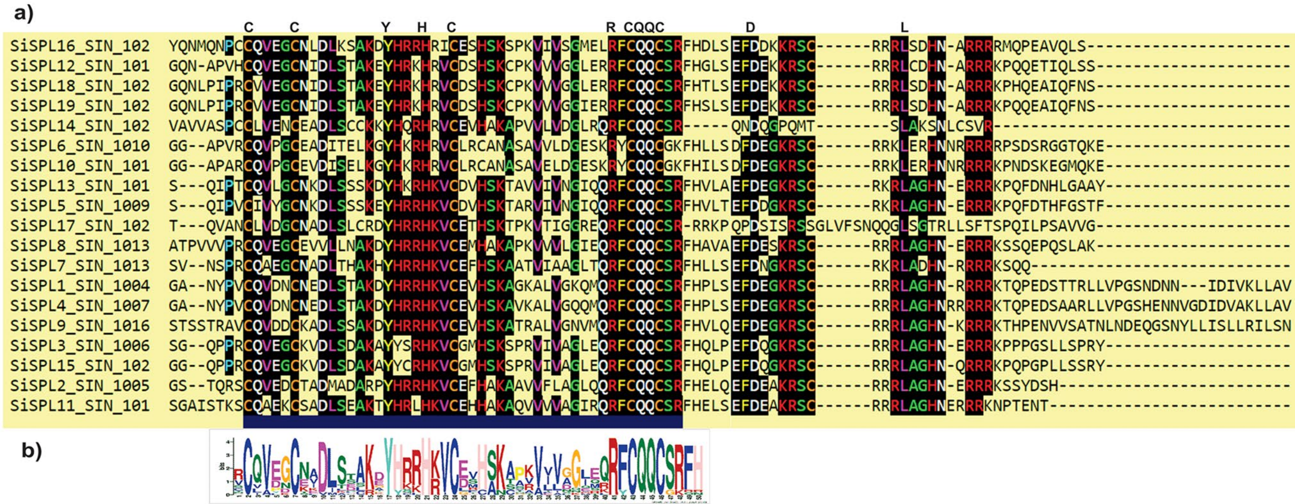

Fig. 6 Multiple protein alignment of SiSPL proteins (**a**). The 'SBP' domain is labeled below the alignment (**b**). The alignment was built using online tool available at http://www.phylogeny.fr/

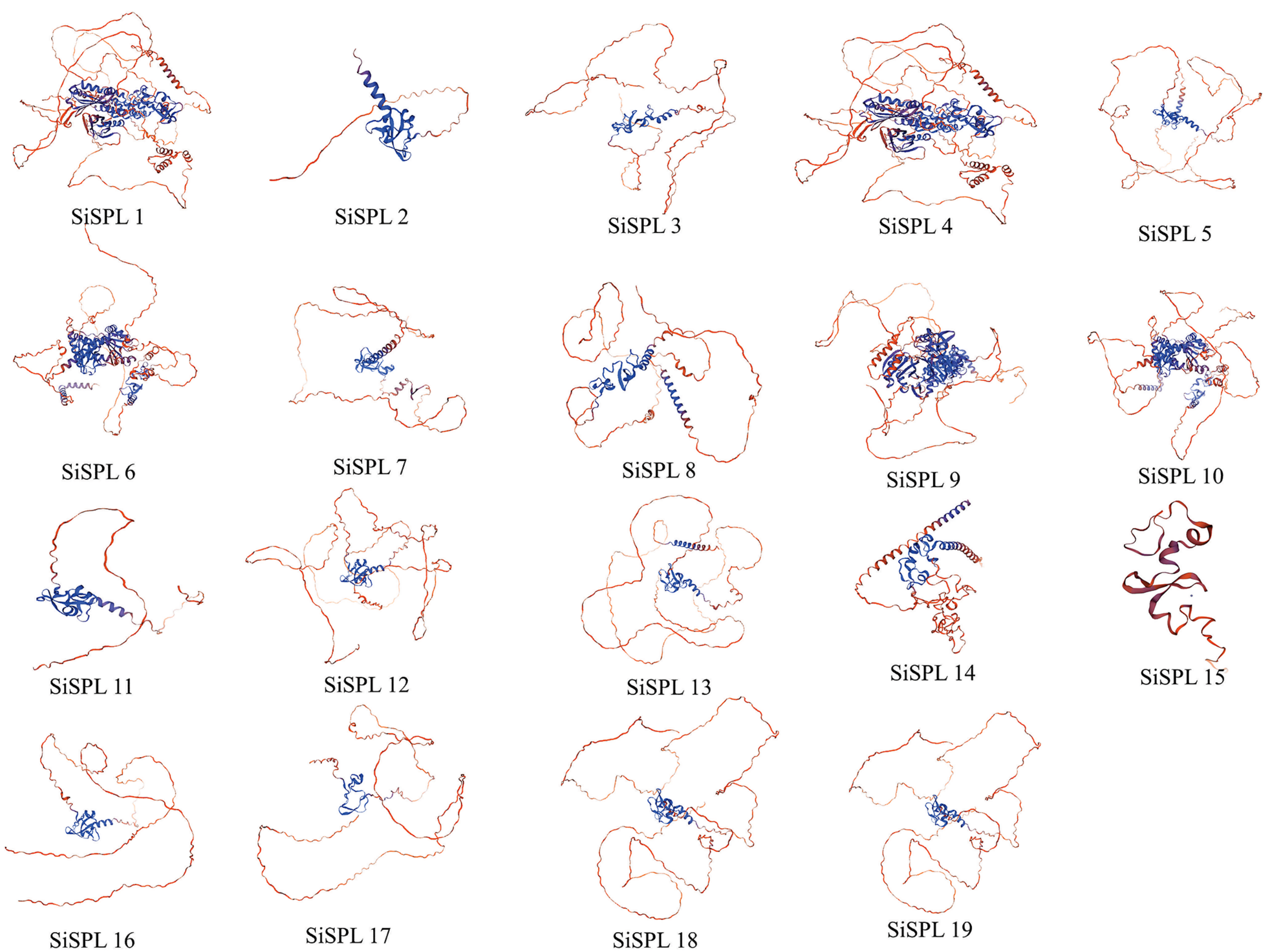

Fig. 7 The 'three-dimensional' (3D) structures of SiSPL proteins

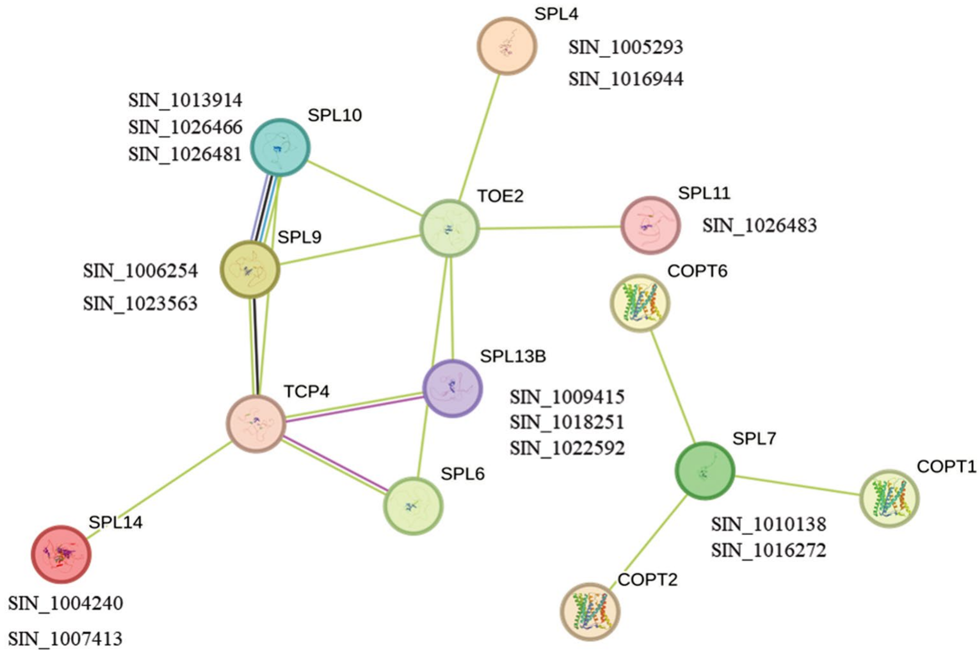

Fig. 8 'Protein-protein interaction' analysis of SiSPL proteins

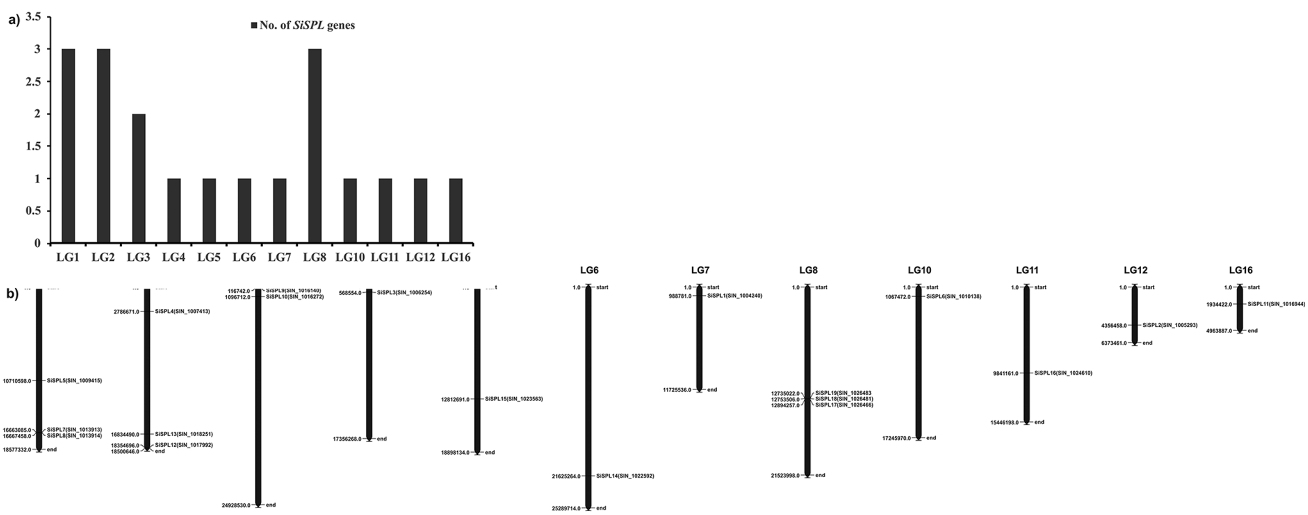

Fig. 9 Genomic location of 19 *SPL* genes in *S. indicum* genome. **a**) Bar chat represented the no. of *SiSPL* genes in each linkage groups (LGs) and **b**) Location of each SiSPL genes in sesame LGs

groups for *Arabidopsis*, 16 genes in 8 groups for tomato and 23 genes in 8 groups for quinoa (Salinas et al. 2012; Schwarz et al. 2008; Zhao et al. 2022).

The gene structure analysis revealed that *SiSPL9* (SIN_1016140) had the maximum (11) while *SiSPL2* (SIN_1005293), *SiSPL11* (SIN_1016944) and *SiSPL17* (SIN_1026466) exhibited lowest (2) number of exons (Fig. 2; Table 1). Likewise, the intron number also differed from 1 to 10 in *SiSPL* genes (Fig. 2). In quinoa, 3–17 exons and 2–16 introns were reported for *SPL* genes (Ren et al. 2022). Likewise, the diverse structure of *SPL* genes was reported in soybean (exons: 1–10 and introns: 0–9) (Tripathi et al. 2017). Also, 1–9 introns have been noticed in *Arabidopsis SPL* genes (Li and Lu 2014). The result indicated that the diverse exon-intron structures of *SiSPL* genes consisted of other plant species, including *Arabidopsis* indicating the conservation of gene structures among these species.

The result of the protein-protein network analysis indicated that SPL7 directly interacts with COPT1, COPT2 and COPT6 proteins. COPTs are copper transporters that are involved in Cu homeostasis, including transportation, absorption and growth in plants (Wang et al. 2021). Moreover, six different SPL proteins (SPL4, SPL6, SPL9, SPL10, SPL11, and SPL13B) have a strong interaction with TOE2 protein. It is evident that TOE2 acts as a transcriptional

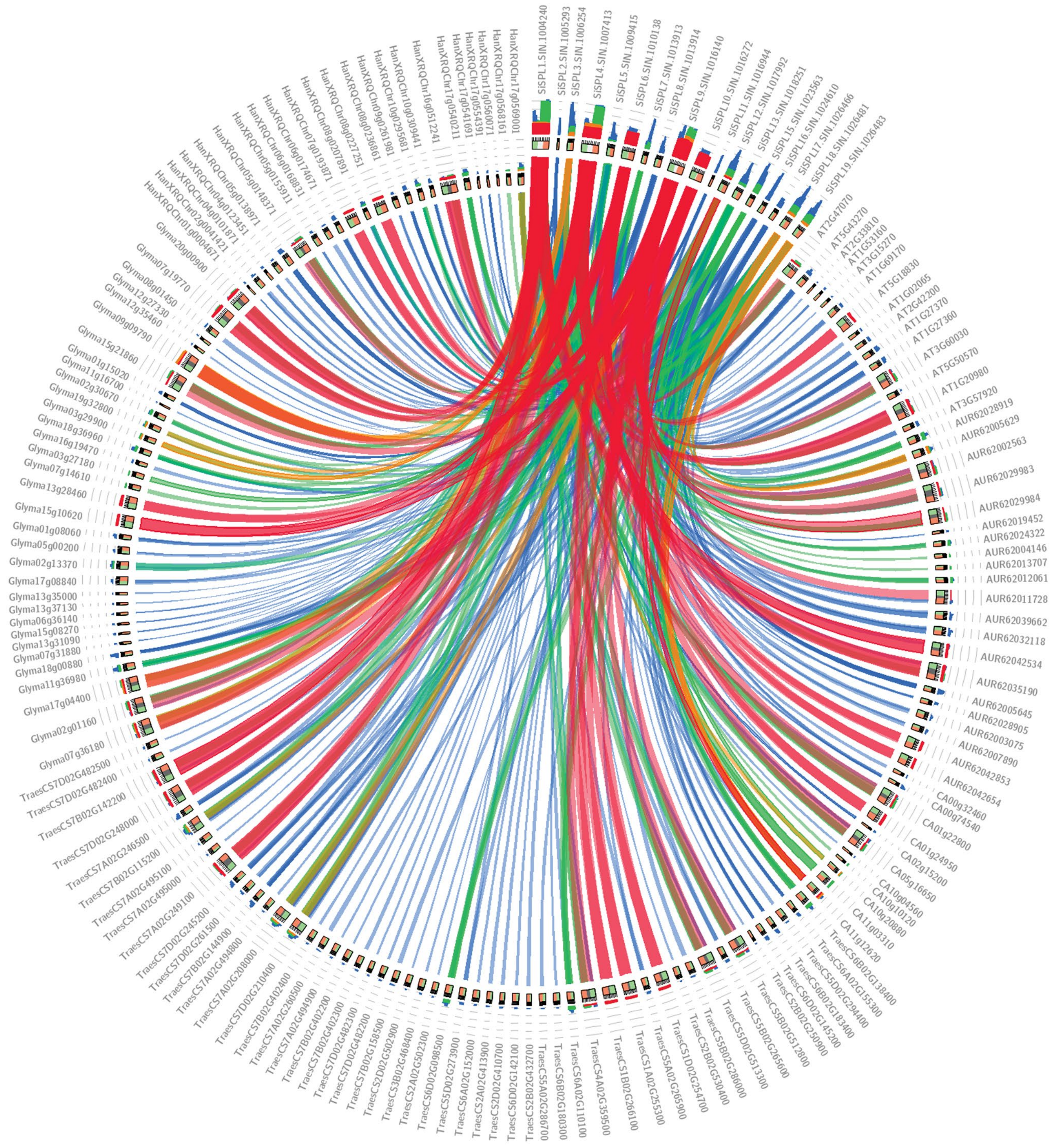

Fig. 10 Syntenic relationship of *SPL* genes among *S. indicum*, *A. thaliana*, *Triticum aestivum*, *Capsicum annum*, *Glycin max and Helianthus annuus* representing the level of conservation at the sequence level. The coloured ribbons indicate the level and intensity of evolutionary conservation among different *SiSPL* genes. The maximum strength is from red to blue

activator and regulates the onset of the main-stem inflorescence trichome in *Arabidopsis* and plays a role in developmental timing (Liu et al. 2023). Additionally, TCP4, a transcription factor involved in ovule and shoot development (Lan et al. 2023; Lin et al. 2016) and has substantial interaction with SPL14, SPL6, and SPL9 proteins. Moreover, two proteins, SPL9 and SPL10, are closely related and share a similar sequence. The outcome of synteny (Fig. 10) and phylogenetic analysis (Fig. 1) exhibited that there could be duplication events in five pairs of *SiSPL* genes (*SiSPL1*

Table 4 The list of five pair duplication events in *SPL* genes of *S. indicum* and their Ka/Ks ratio

Gene 1	Gene 2	Non-synonymous substitution (Ka)	Synonymous substitution (Ks)	Ka/Ks
SIN_1004240	SIN_1007413	0.0743	0.3130	0.2374
SIN_1006254	SIN_1023563	0.1220	0.5012	0.2434
SIN_1009415	SIN_1018251	0.2875	0.8649	0.3324
SIN_1010138	SIN_1016272	0.1688	0.6749	0.2502
SIN_1026481	SIN_1026483	0.0520	0.1605	0.3239

and *SiSPL4*, *SiSPL3* and *SiSPL15*, *SiSPL5* and *SiSPL13*, *SiSPL6* and *SiSPL10*, and *SiSPL18* and *SiSPL19*). Further, the 'all-vs-all' BLASTP (threshold e-value less than 1 × e^{-152}) The gene duplication analysis (Ks/Ks ratio) of *SiSPL* gene pairs was less than 1.0 (0.2374 to 0.3324), which revealed that there is a robust purifying selection force in the evolution of *SiSPL* proteins. Similar to our results, the strong purifying selection pressure was also reported in cotton, peanut and *Salvia miltiorrhiza* for the evolution of WRKY proteins (Ding et al. 2014; Li et al. 2015; Song et al. 2016). In the promoter region of 19 *SiSPL* genes, a total of 454 stress responsive '*cis*-regulatory' elements were identified (Table 5), revealing their involvement in stress responses. Similar cis-acting elements associated with stress responses have been reported for *SPL* genes in quinoa (Ren et al. 2022). The GO enrichment of 19 SiSPL proteins revealed that 'GO:0010321' and 'GO:0010229' were the most enriched terms for BP category that were involved in the regulation of vegetative phase change and inflorescence development, respectively (Fig. 11). It has been reported that three *SPL* genes (*SPL3/4/5*) stimulate transition of the vegetative phase as well as flowering in *Arabidopsis* (Wu and Poethig 2006).

Of 19 *SiSPLs*, seven genes showed SSR motifs ranging from mono- to tri-nucleotides (Table 6). Zeng et al. (Zeng et al. 2019) and Yu et al. (Yu et al. 2020) also found similar SSR variants in *Citrus clementina* and *Jatropha curcas*, respectively. The RNA-seq based expression data showed that the transcript level of *SiSPL9* (SIN_1016140) followed by *SiSPL12* (SIN_1017992) and *SiSPL3* (SIN_1006254) genes was higher in PEG-induced drought conditions than the control roots while the opposite result was found for *SiSPL4* (SIN_1007413) and *SiSPL17* (SIN_1026466) in sesame (Fig. 12b). The results suggest that these genes might be drought responsive. Similar to this study, Mao et al. (Mao et al. 2016) reported the transcript abundance of *ZmSPL2* and *ZmSPL20* in cold and drought treated samples in maize. Wang et al. (Wang et al. 2023) also reported the drought stress responsiveness of *SPL* genes in alfalfa. Moreover, the outcome of the filtering of 19 *SiSPL* gene transcriptomic data as previously reported by (Zhang et al. 2020) under salinity stress in sesame revealed that the expression of *SiSPL8* (SIN_1013914), *SiSPL15* (SIN_1023563) and *SiSPL17* (SIN_1026466) was up-regulated in both genotypes at both time-points (Fig. 12c). Mao et al. (Mao et al. 2016) also reported that *SPL* genes (*ZmSPL10*) are induced by salinity. In addition, Ning et al. (Ning et al. 2017) reported that *BpSPL9* is associated with drought and salinity tolerance in birch (*Betula platyphylla*). It has been reported that overexpression of *MsSPL8* exhibited enhanced drought and salinity tolerance in transgenic plants of alfalfa (Gou et al. 2018). The FPKM expression of *SiSPL1* and *SiSPL9* genes was higher in the root with waterlogging conditions compared to the control roots at flowering stage (Fig. 12d) which revealed their possible involvement in waterlogging stress. Ren et al. (Ren et al. 2022) showed that the expression of *CqSPL1* and *CqSPL5* was up-regulated, whereas the expression of *CqSPL2* was downregulated in waterlogging stress. Lai et al. (Lai et al. 2022) reported that three *SPL* (*SPL9*, *SPL10* and *SPL16)* genes were the highly expressed in various tissues (root, stem, and leaf) under multiple stresses (acid, alkali, NaCl, PEG, and flooding stress). However, further qRT-PCR validation is required under various stress treatments and control samples in sesame to support these findings.

Table 5 *Cis*-acting regulatory elements in the promoter regions of *SiSPL* genes

	Gene name	*SiSPL1*	*SiSPL2*	*SiSPL3*	*SiSPL4*	*SiSPL5*	*SiSPL6*	*SiSPL7*	*SiSPL8*	*SiSPL9*	*SiSPL10*	*SiSPL11*	*SiSPL12*	*SiSPL13*	*SiSPL14*	*SiSPL15*	*SiSPL16*	*SiSPL17*	*SiSPL18*	*SiSPL19*	Total	
Light response	AE-BOX	0	1	0	0	1	0	0	0	1	0	0	1	0	0	2	0	1	0	0	**7**	
	BOX-4	7	4	9	2	5	4	9	7	7	13	4	2	11	6	3	3	1	6	1	**104**	
	G-box	4	10	0	4	1	2	1	0	3	6	2	0	0	2	8	2	0	4	3	**52**	
	GA-motif	1	0	0	0	1	0	2	0	0	1	0	0	0	0	0	0	1	1	0	**7**	
	GATA-motif	1	1	0	5	4	1	0	2	2	0	0	1	3	0	1	0	1	0	2	**24**	
	I-box	1	0	0	2	4	1	0	0	1	1	0	1	0	0	0	2	0	0	0	**13**	
	MRE	0	0	1	1	0	0	1	1	0	1	1	1	0	1	0	0	1	0	0	**9**	
	GT1-motif	1	4	1	1	0	2	6	2	5	6	4	1	0	4	1	1	5	3	2	**49**	
	Sp1	0	0	0	1	0	3	0	0	1	2	0	0	0	0	0	0	0	0	0	**7**	
	ATCT-motif	0	0	0	1	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	**3**	
	ACE	0	0	0	0	0	2	0	0	1	0	0	1	0	2	0	0	0	0	0	**6**	**281**
Phytohormonhe response	ABRE	5	10	0	4	1	3	1	1	4	5	2	0	0	2	6	2	0	4	2	**52**	
	GARE-motif	0	0	0	1	1	0	0	0	1	0	0	2	0	1	1	1	0	0	0	**8**	
	ERE	2	2	2	0	3	5	5	2	3	3	4	1	4	2	2	1	3	3	3	**50**	
	TATC-box	0	0	0	0	0	0	0	0	1	0	0	0	1	0	1	1	0	0	0	**4**	
	P box	0	0	0	1	0	0	1	1	2	0	0	0	0	0	0	1	2	0	0	**8**	
	TCA-element	0	2	1	1	0	0	0	0	4	1	0	1	0	1	1	2	1	0	1	**16**	
	TGA-element	0	1	0	1	1	1	0	2	0	2	3	0	0	1	0	1	1	3	1	**18**	
	TGACG-motif	1	1	1	1	1	2	1	3	0	5	2	0	1	0	0	2	0	3	3	**27**	**183**
Stress response	ARE	0	3	2	2	2	2	3	3	3	4	2	4	2	2	5	1	3	3	4	**50**	
	box S	1	1	1	0	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0	**5**	
	DRE C0RE	0	2	0	0	1	0	0	1	0	0	0	0	0	1	1	1	0	0	0	**7**	
	MBS	1	0	0	4	2	2	2	5	0	1	0	1	3	0	1	0	0	2	2	**26**	
	STRE	6	4	4	3	2	2	6	5	5	5	5	5	5	1	4	4	6	2	3	**77**	
	TC-rich repeats	0	0	0	0	0	4	0	0	1	2	0	0	1	1	0	1	2	0	0	**12**	
	W-box	2	1	1	0	2	0	0	3	1	0	0	1	2	2	0	0	0	0	0	**15**	
	WRE3	0	0	1	0	1	2	2	1	3	1	1	0	2	1	0	2	1	0	0	**18**	
	WUN-motif	2	0	3	1	1	2	2	0	1	0	2	1	3	2	2	3	1	0	1	**27**	
	CCAAT-box	1	0	0	0	0	0	0	1	1	0	1	0	0	0	0	0	0	0	0	**4**	
	MYB	1	10	2	5	8	3	1	5	5	9	4	4	3	4	8	7	4	5	5	**93**	
	MYB recognition site	1	0	0	0	0	0	0	1	1	0	1	0	0	0	0	0	0	0	0	**4**	
	Myb-binding site	1	2	0	2	4	1	1	1	2	2	1	2	1	1	1	3	0	0	1	**26**	
	MYC	2	0	7	4	2	5	3	3	9	5	6	2	5	7	6	3	2	2	2	**75**	
	LTR	0	1	0	2	0	0	0	0	1	0	0	3	0	0	2	4	0	1	1	**15**	**454**
Total no. of motifs		**41**	**60**	**36**	**49**	**48**	**50**	**47**	**51**	**69**	**75**	**45**	**36**	**47**	**45**	**56**	**48**	**36**	**42**	**37**	**918**	

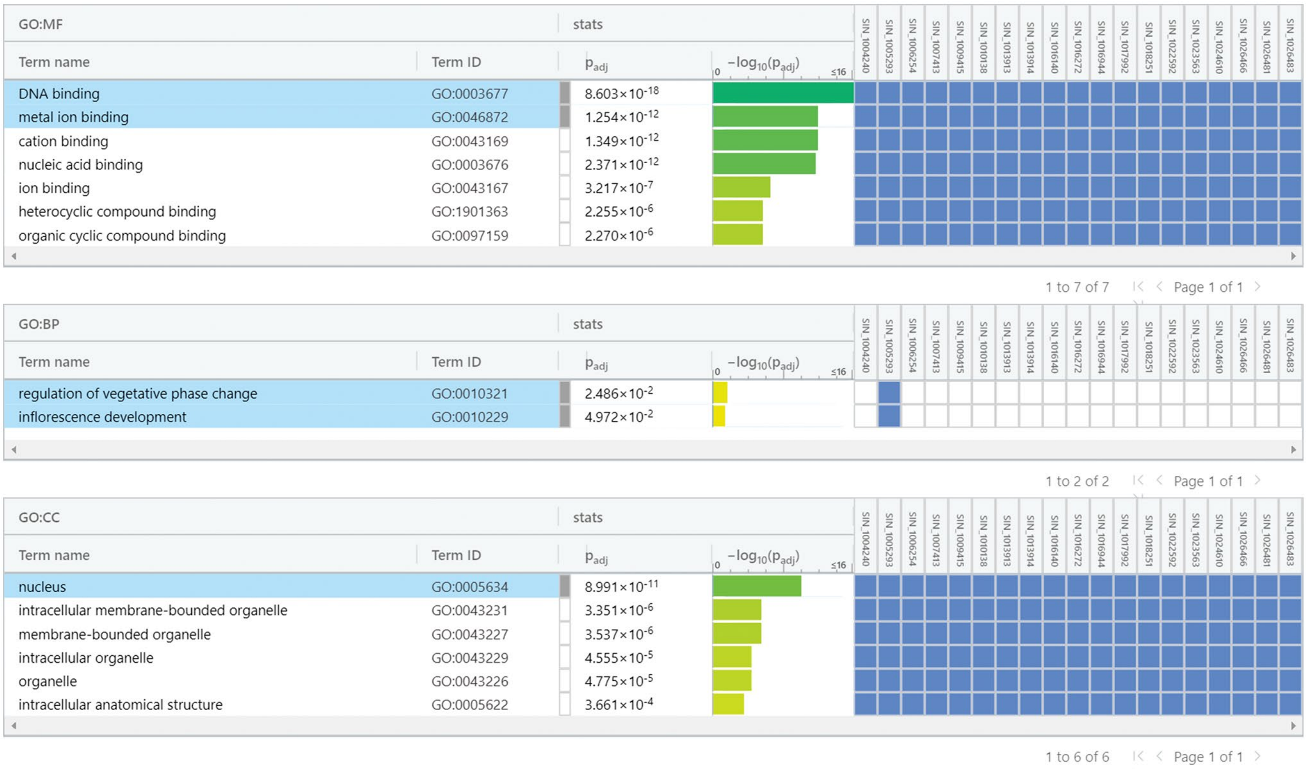

Fig. 11 GO enrichment analysis for 19 *SiSPL* genes

Table 6 The SSR present in the *SiSPL* genes

Gene name	Gene ID	SSR nr.	SSR type	SSR	Size (bp)	Start	End
SiSPL1	SIN_1004240	1	p1	(T)11	11	2239	2249
SiSPL1	SIN_1004240	2	p2	(TA)8	16	2406	2421
SiSPL4	SIN_1007413	1	p1	(T)17	17	1714	1730
SiSPL4	SIN_1007413	2	p2	(TA)7	14	2797	2810
SiSPL4	SIN_1007413	3	p3	(AGG)5	15	3049	3063
SiSPL6	SIN_1010138	1	p3	(CTT)7	21	935	955
SiSPL6	SIN_1010138	2	p1	(T)10	10	959	968
SiSPL6	SIN_1010138	3	p2	(TC)9	18	1062	1079
SiSPL9	SIN_1016140	1	p1	(T)19	19	606	624
SiSPL14	SIN_1022592	1	p3	(GAA)7	21	64	84
SiSPL15	SIN_1023563	1	p1	(T)12	12	368	379
SiSPL15	SIN_1023563	2	p1	(T)10	10	1255	1264
SiSPL17	SIN_1026466	1	p3	(CAG)5	15	1645	1659

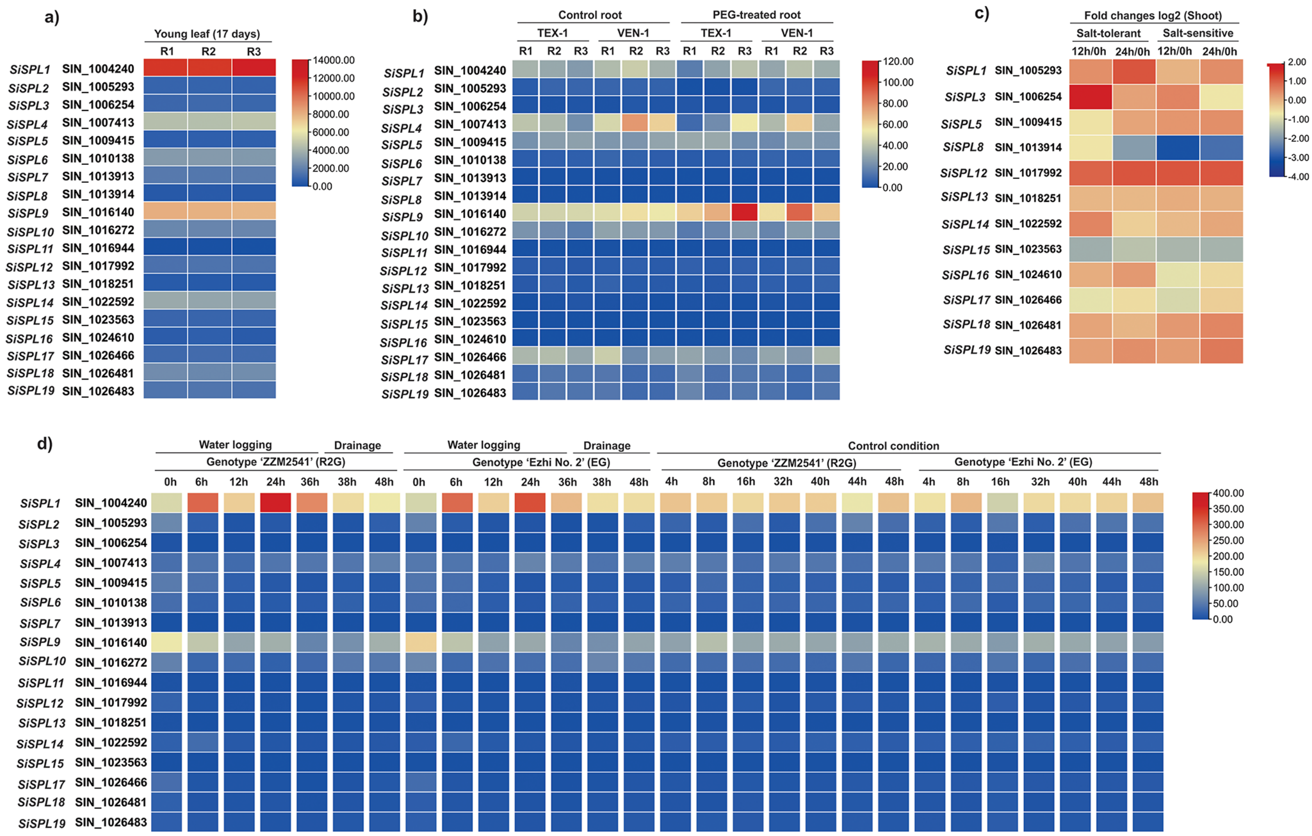

Fig. 12 Heatmap representation of the FMKM/RPKM expression pattern of *SiSPL* genes in *S. indicum*. The FMKM value was retrieved from GEO dataset of the NCBI database. (**a**) Expression pattern of young leaf (GSE164511), (**b**) control and PEG-treated root (GSE148340 of *S. indicum*) and (**c**) RPKM value of salt-tolerant and salt-sensitive shoots of *S. indicum* under salt-stress condition (Zhang et al. 2020), and (**d**) FPKM value of water logging and control root (GSE133186) of *S. indicum* at flowering stage (Dossa et al. 2019)

Conclusion

In this study, we identified a total of 19 *SiSPL* genes and their physicochemical characteristics, location in the genome, subcellular localization, exon-intron structure, conserved domains and motifs, *cis*-regulatory elements, protein structure, RNA-seq based expression in sesame were analyzed. The available sesame transcriptomic data (FPKM/RPKM) showed that *SiSPL* genes varied in their expression between stressed vs. control samples. *SiSPL9* (SIN_1016140), *SiSPL12* (SIN_1017992) and *SiSPL3* (SIN_1006254) might be drought responsive, whereas *SiSPL8* (SIN_1013914), *SiSPL15* (SIN_1023563) and *SiSPL17* (SIN_1026466) could be salinity responsive. Besides, *SiSPL1* and *SiSPL9* genes might be waterlogging responsive. These *SiSPL* genes can be potential candidates for drought, salinity and waterlogging stress tolerance in *S. indicum*.

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Data availability The data supporting the results of this manuscript are presented in the article/supplementary materials.

Declarations

Conflict of interest The authors declare they have no competing interests.

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